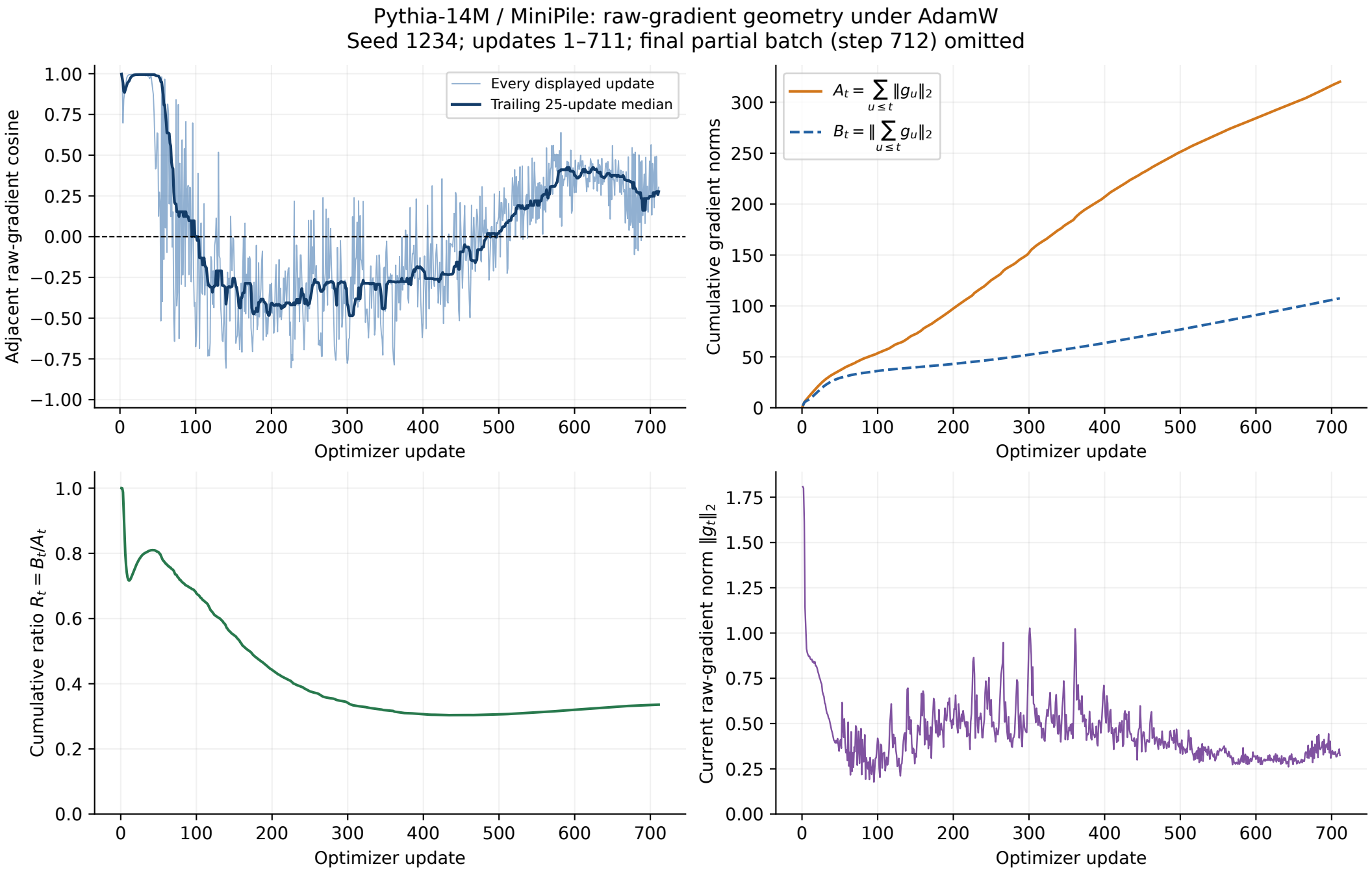

Pythia-14M / MiniPile: raw-gradient geometry under AdamW
Seed 1234; updates 1–711; final partial batch (step 712) omitted
Adjacent raw-gradient cosine
Every displayed update
Trailing 25-update median
Optimizer update
Cumulative gradient norms
$A_t = \sum_{u \le t} \|g_u\|_2$
$B_t = \| \sum_{u \le t} g_u \|_2$
Optimizer update
Cumulative ratio $R_t = B_t / A_t$
Optimizer update
Current raw-gradient norm $\|g_t\|_2$
Optimizer update